Project 2: R&D Investment under National Uncertainty

DATA 201 - Spring 2026

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Motivation

Future investments in R&D depend on wealth, human development, policy, competitiveness, and may be influenced by uncertainty factors.

QUESTIONS:

1. Can we quantify how wealth drives technology (GDP/capita vs. R&D expenditure)?

2. Have different regions of the world developed independent correlations?

3. How influential are human factors (health and education) on R&D expenditure?

4. Does uncertainty affect R&D expenditure?

Datasets

World uncertainty index (WUI)

http://worlduncertaintyindex.com/data/

(tracks uncertainty (sentiment analysis) quarterly across the globe since 1950s. 143 countries.)

World Bank - Development Indicators (WBI)

https://databank.worldbank.org/source/world-development-indicators

(6 indicators - timeperiod: 1995-2024, 52 selected countries.)

Bias: sparse information for the less developed or more restrictive countries.

Although a good crossection of WUI and WBI was identified, the 52 countries do not include any of the low-income nations.

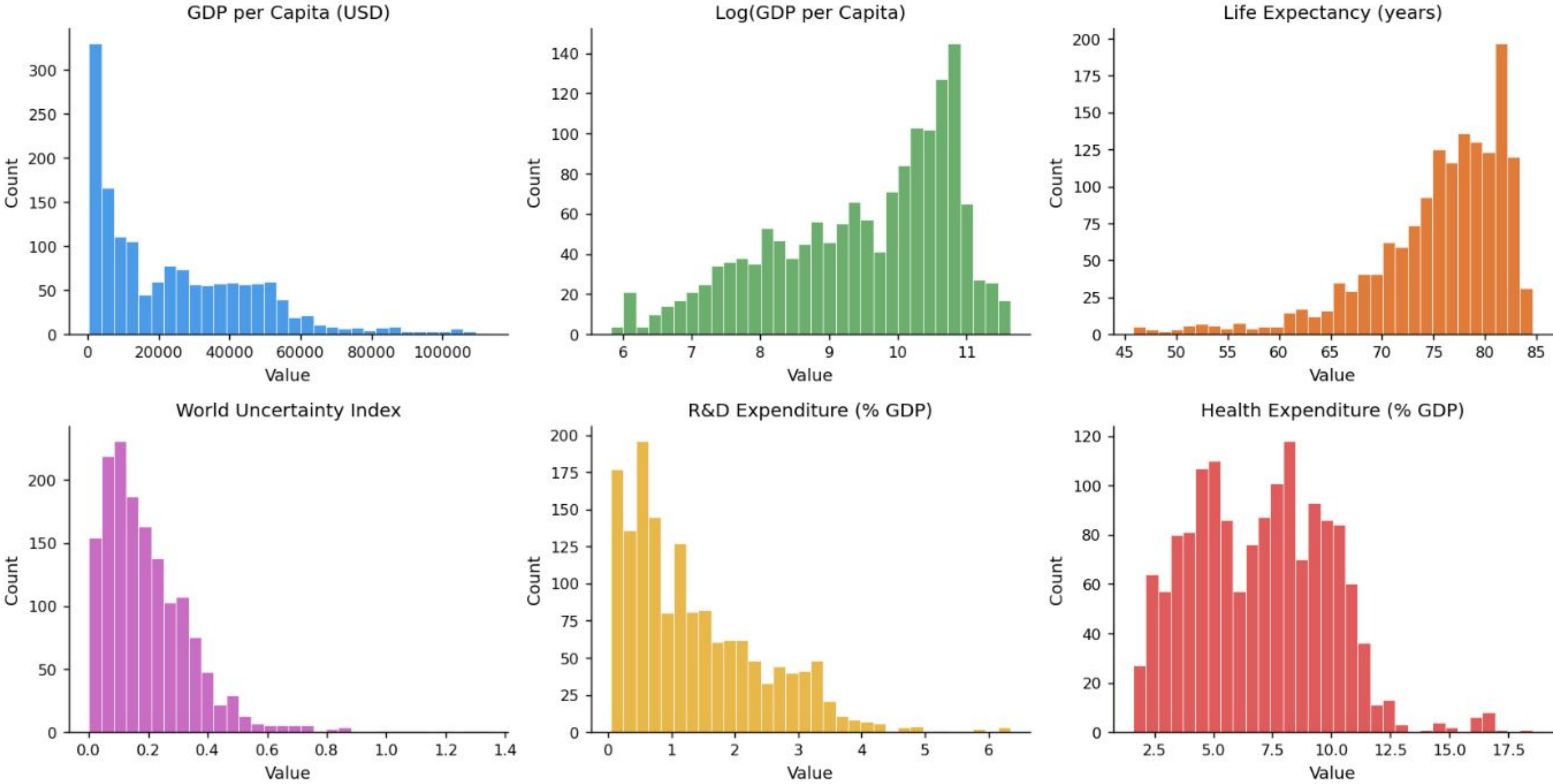
Merged working dataset after cleaning:

1560 rows and 11 variables

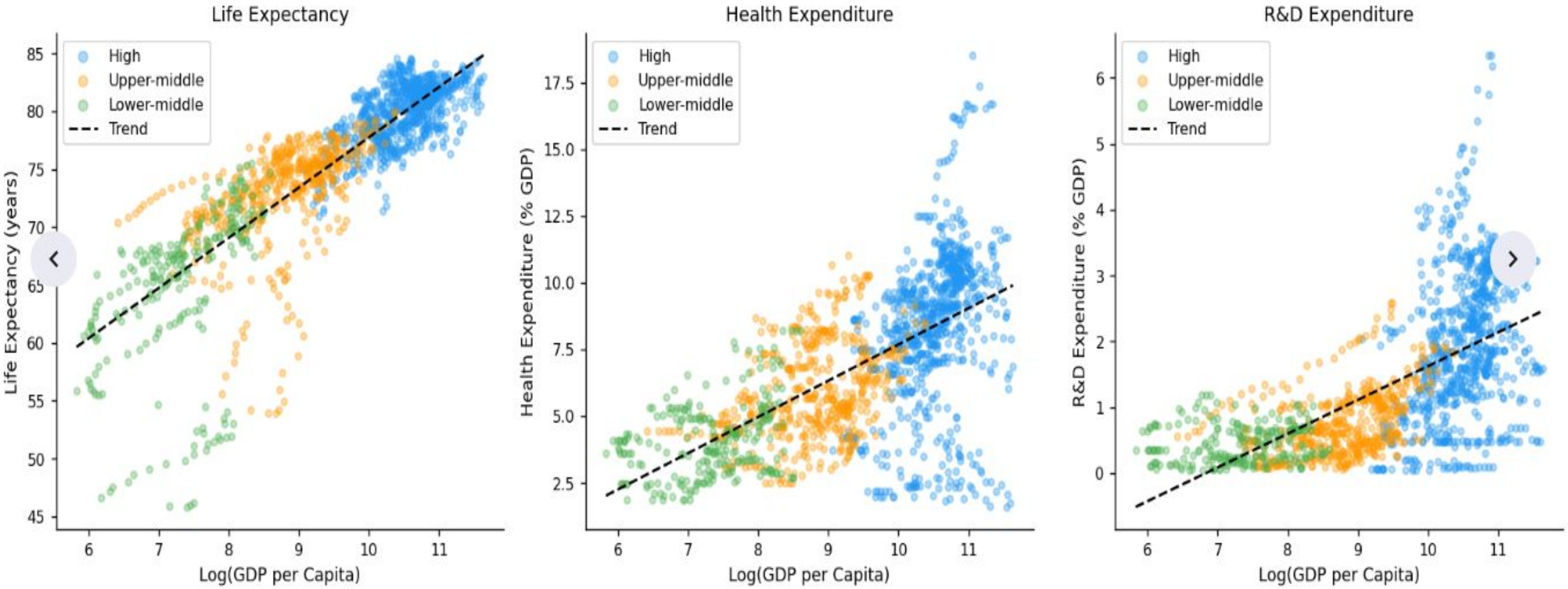
Variables of interest

Variable		Type
iso_code	ISO 3-letter country code	Categorical
region	middle-east, latin-america, Asia-pacific, europe, north-america	Categorical
income_group	lower-middle, upper-middle, high	Categorical
year		Quantitative
GDP_per_capita	US dollars	Quantitative
Life_expectancy	years	Quantitative
Health_spend	%GDP	Quantitative
Education_spend	%GDP	Quantitative
RD_spend	%GDP	Quantitative
WUI	uncertainty index	Quantitative

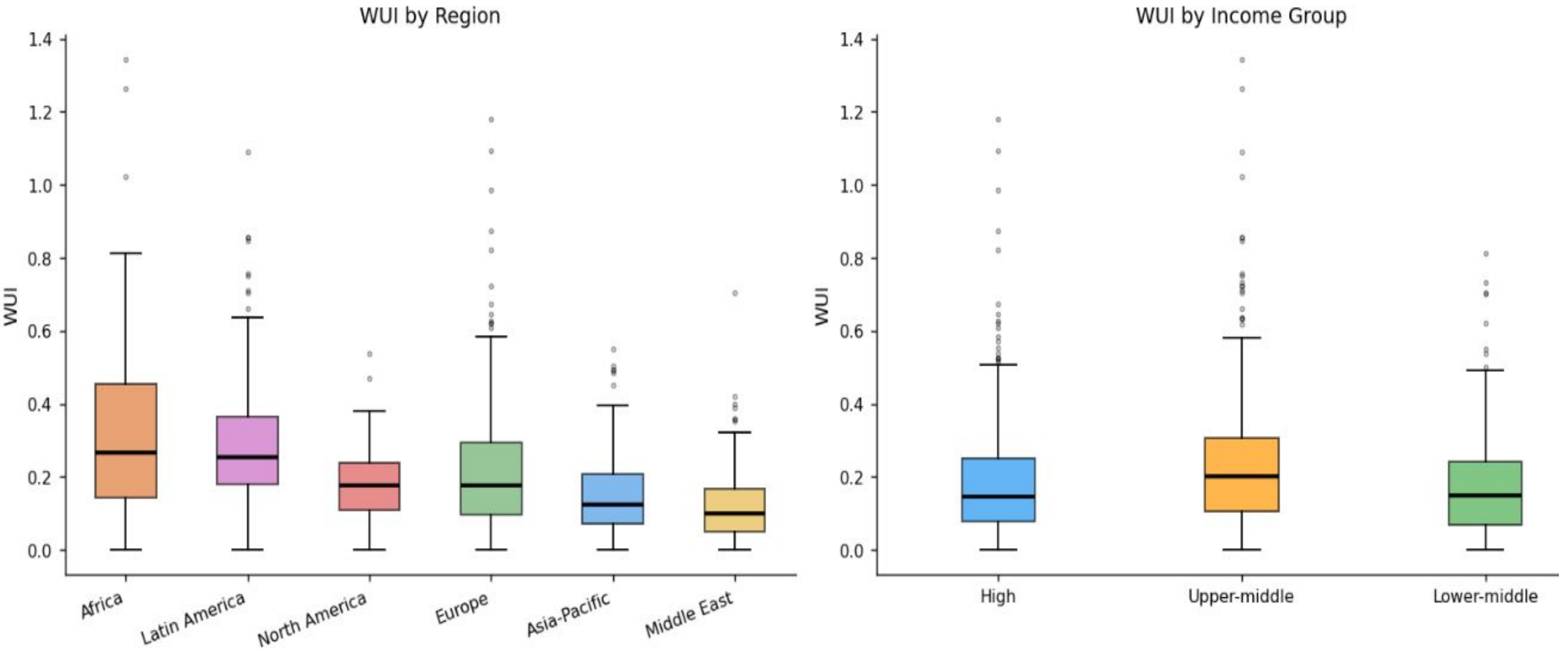
Exploratory data analysis and visualization



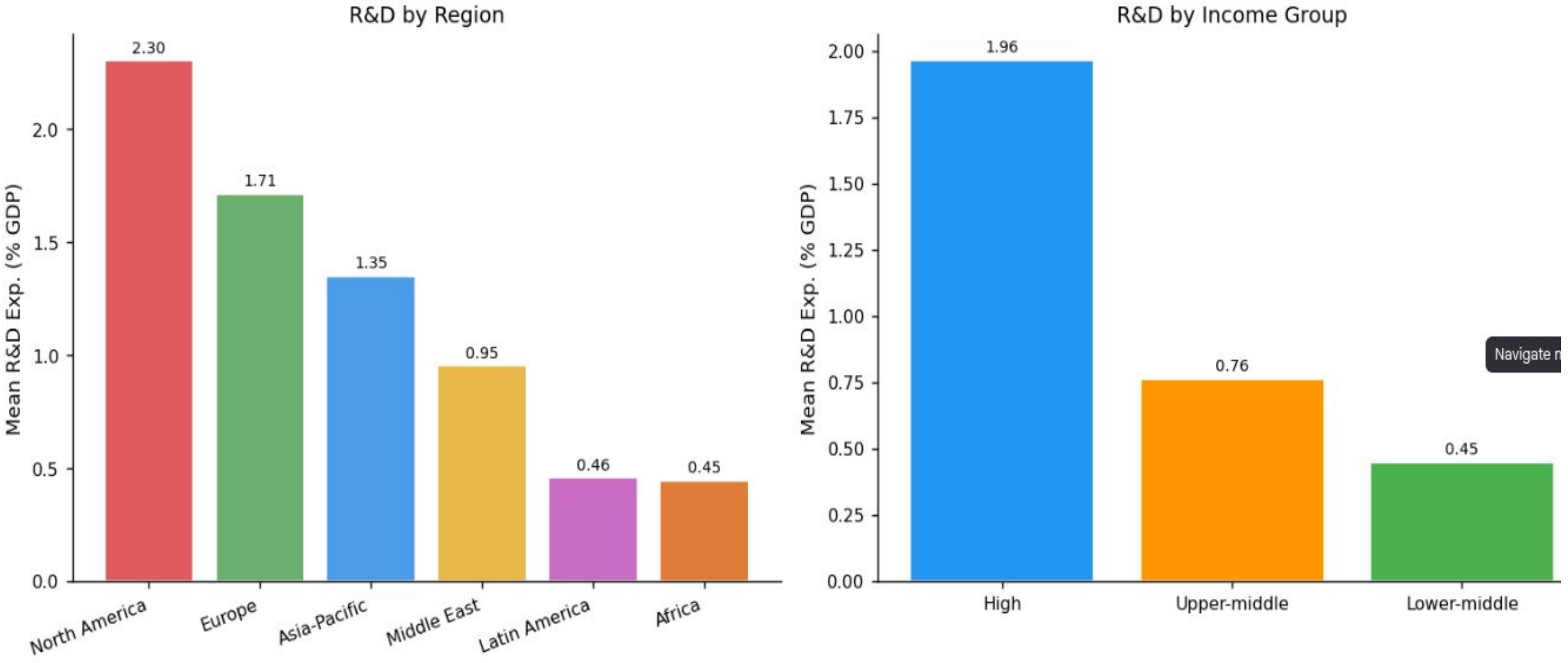
Correlations with respect to log_gdp/capita



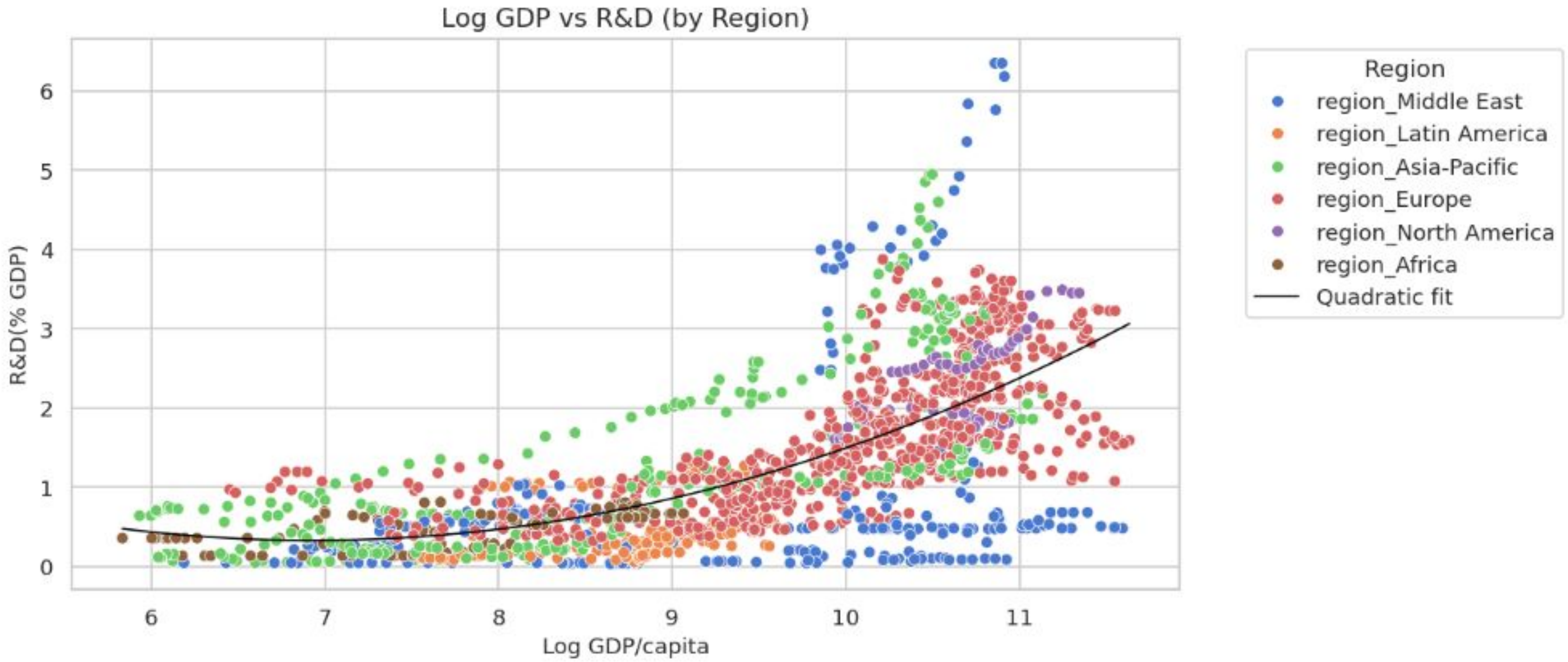
World Uncertainty Index by region and income group

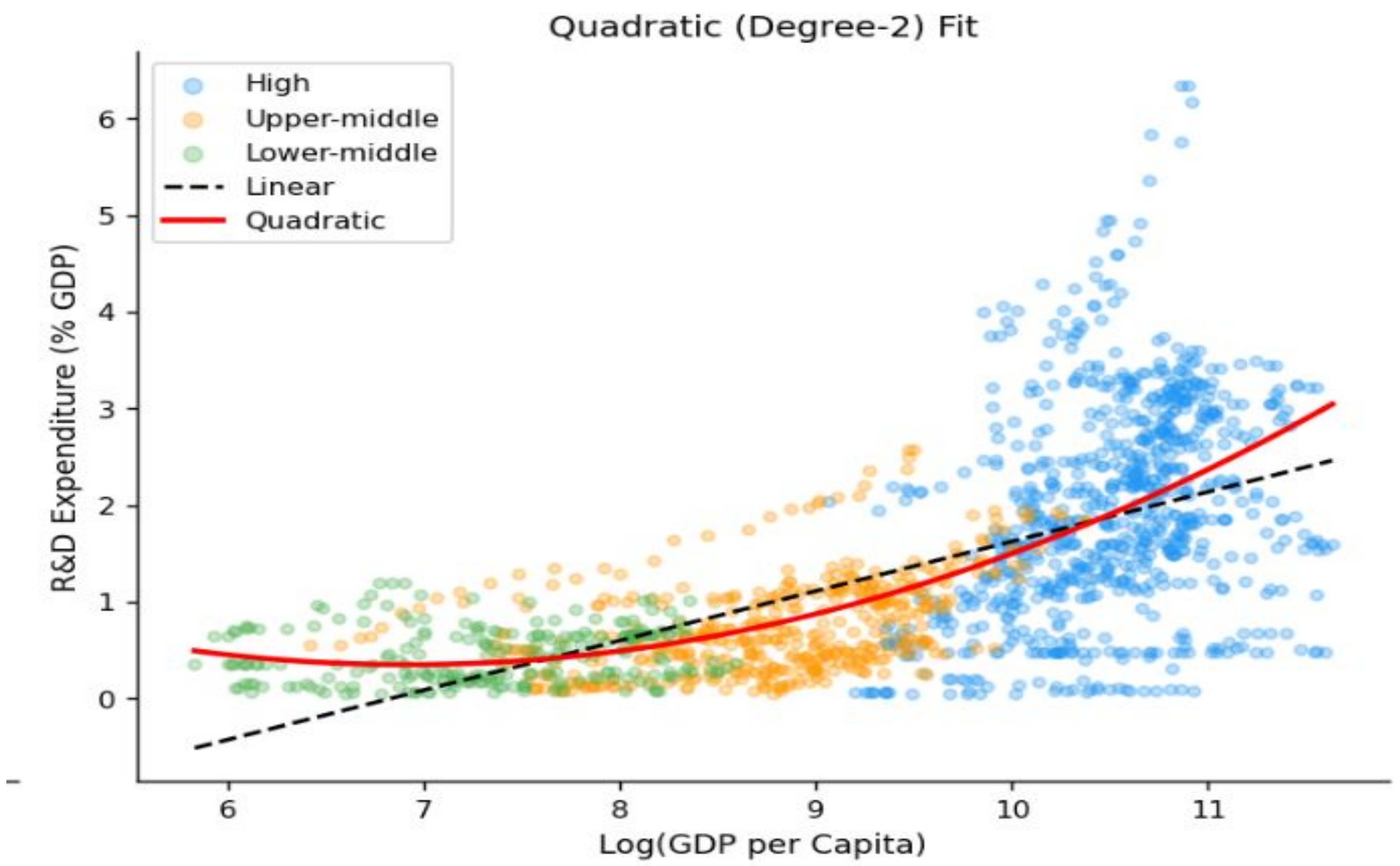


Mean R&D investment by region and income group



Quadratic fit to R&D vs log GDP/capita




Quadratic (Degree-2) Fit
High
Upper-middle
Lower-middle
Linear
Quadratic
R&D Expenditure (% GDP)
0
1
2
3
4
5
6
6
7
8
9
10
11
Log(GDP per Capita)

Models Results for prediction of R&D

Tried 4 models:

0. Linear Regression without region and income group included
1. Linear regression including all features, including region and income group
2. Polynomial regression: quadratic in log GDP
3. Interaction of log GDP and WUI

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Model Comparison:
                     Model      RMSE        R2
0                   Linear  0.780412  0.506989
1  Linear with Categorical  0.688461  0.616321
2               Polynomial  0.689879  0.614739
3              Interaction  0.688766  0.615981
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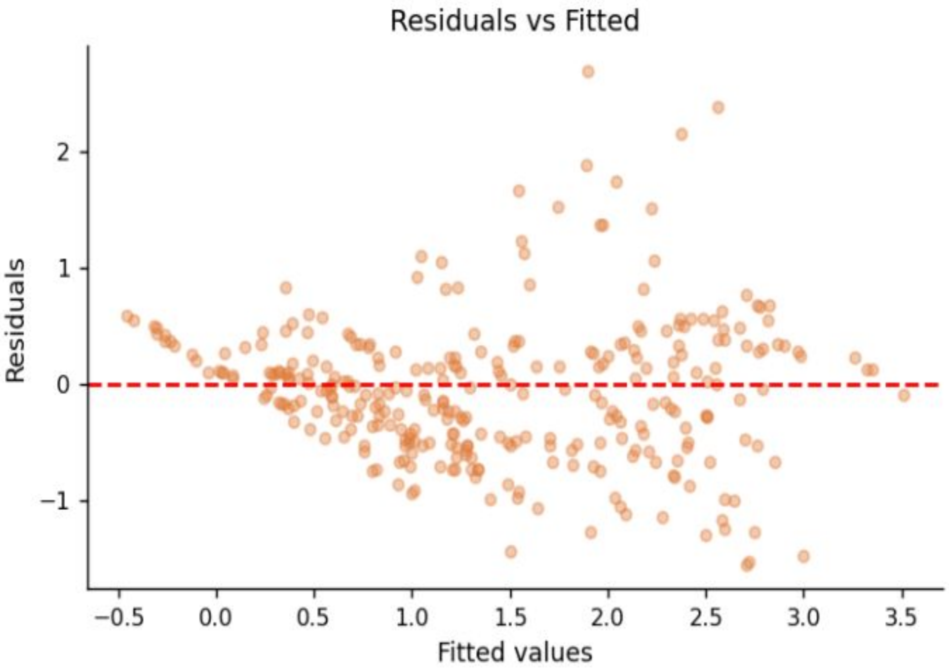
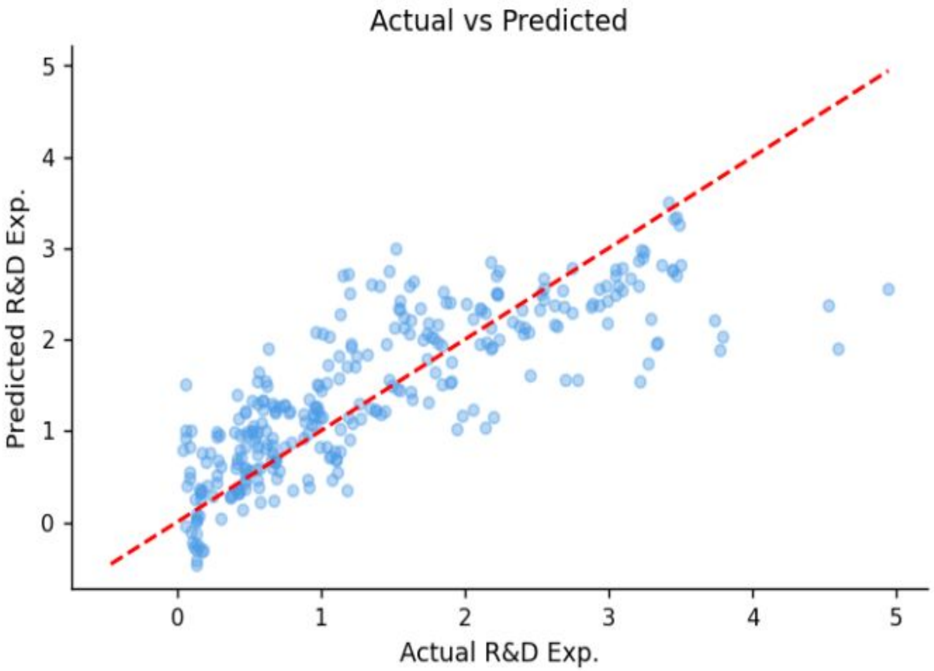
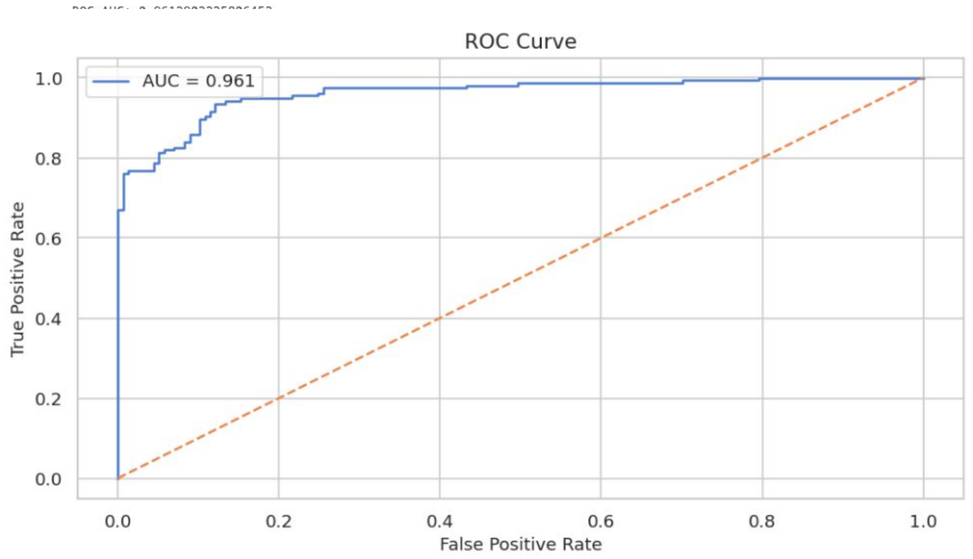
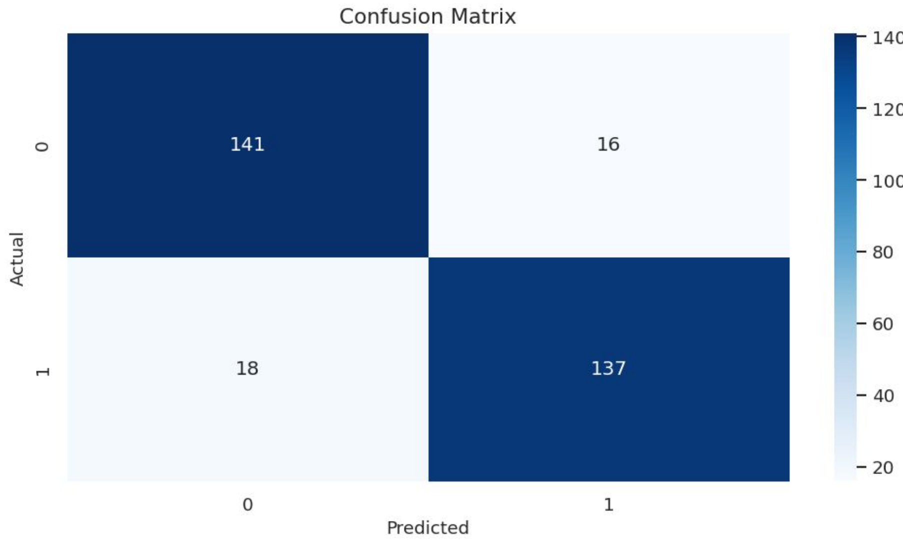
Model 2: Residuals and Actual vs Predicted R&D (% GDP)





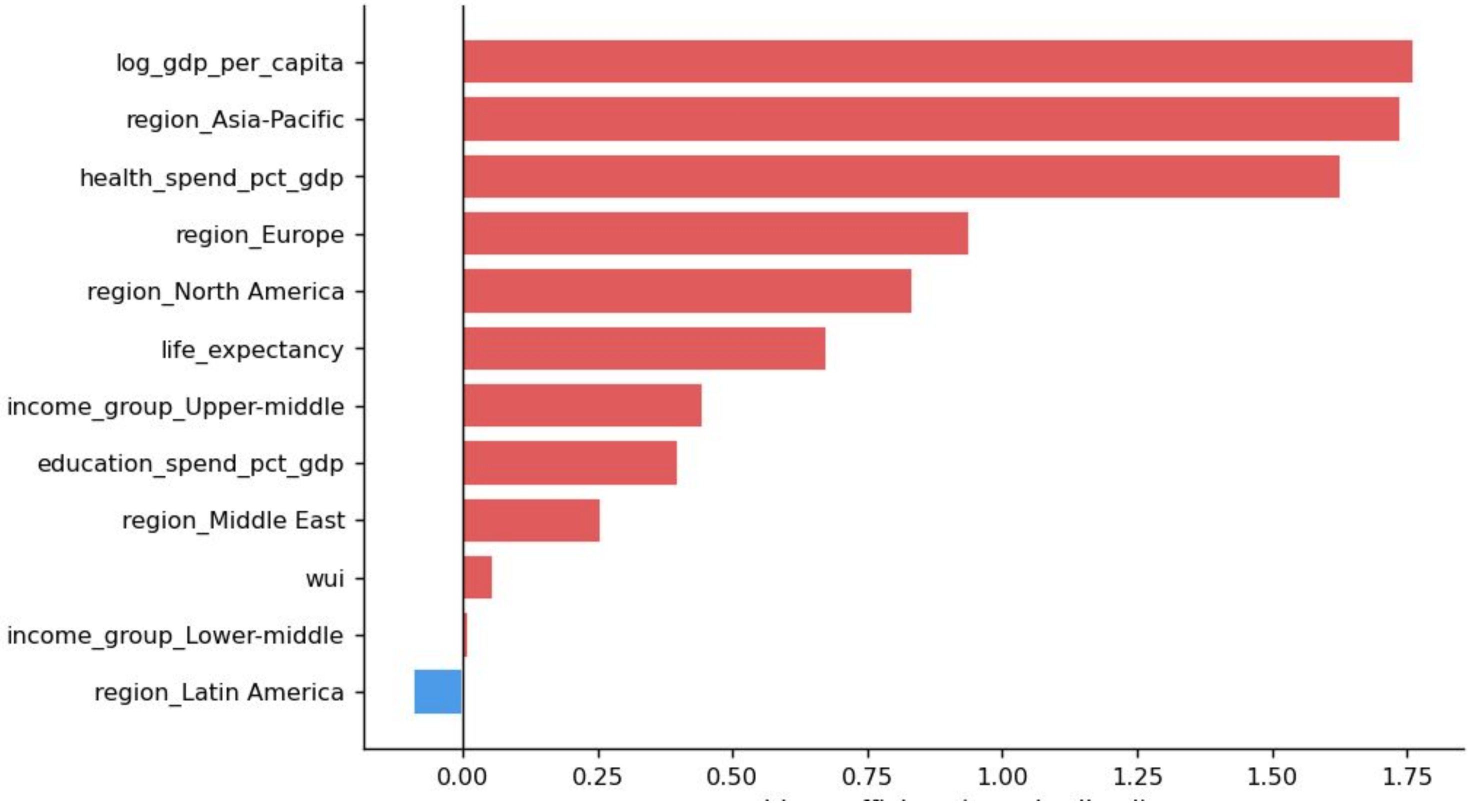
Logistic regression and classification of countries by R&D expenditure





Regression accuracy: 0.89

Logistic correlation coefficients vs log odds: variable importance



Conclusions

R&D expenditure across countries, focusing on the roles of economic development, human development, and uncertainty:
First, economic wealth is the dominant driver of R&D investment. Diminishing returns.

Second, structural factors such as region and income group play a critical role.

Third, the role of uncertainty appears to be limited.One possible explanation is that R&D expenditure reflects long-term strategic investment decisions. Structural factors are more important determinants.

These insights have implications for policy, as they emphasize the importance of sustained economic development and institutional support for innovation, over short-term responses to uncertainty. I.e., continue to fund good research even during tough times!